FEDERAL SPECIFICATION

CANNULA, BRAIN, FRAZIER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers brain cannula, Frazier design, for use during neurosurgical operative procedures.

1.2 Classification.

1.2.1 *Types.*—Cannulae shall be of the following types, as specified (see 6.1).

Type I. —Exploring cannula set (set of three (3)).

Type II. —Ventricular cannula.

Type III.—Suction tube.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG-I-526 — Instruments, Dental and Surgical; General Specifications.
QQ-B-626—Brass, Leaded and Non-Leaded; Rods, Shapes, Forgings, and Flat Products with Finished Edges (Bars, Flat Wire and Strips).
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking For Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-B-117—Bags, Interior, Packaging.
MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the

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procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply:

American Standards Association, Inc., Standards:

Z70.1–1955–(EDC615.47) — American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American Standards Association, Incorporated, 70 East Forty-fifth Street, New York 17, N. Y.).

3. REQUIREMENTS

3.1 **Material.**

3.1.1 *Type I.*—Type I cannula shall be fabricated of class 4, composition T metal, conforming to Federal Specification GG–I–526.

3.1.2 *Type II.*—Type II cannula shall be fabricated of class 2, composition G(c) steel, conforming to Federal Specification GG–I–526.

3.1.3 *Type III.*—Type III cannula shall be fabricated either of class 4, composition N metal, conforming to Federal Specification GG–I–526, or of composition 22 of Federal Specification QQ–B–626, chromium plated over nickel. Plating requirements and corrosion tests requirements shall be in accordance with Federal Specification GG–I–526 for both materials.

3.2 **Style, design, and dimensions.**—All cannulae hereinafter specified shall be suitable for use during neurosurgical operative procedures.

3.2.1 *Type I (exploring cannula set).*—Type I shall consist of a set of three brain exploring cannulae, Frazier design, one small, one medium, and one large size. The cannula shall be in accordance with figure 1.

3.2.2 *Type II (ventricular).*—Type II shall be a 10 Fr. (3.33 mm.) straight brain cannula with stylet, Frazier design, with a dull point and a standard Luer hub cavity (see 3.2.2.1). The cannula shall be in accordance with figure 2 and shall be marked in millimeters as shown in the figure.

3.2.2.1 *Hub cavity.*—Hub cavity shall be a standard Luer hub cavity in accordance with figure 3 of American Standards Association (ASA) Standard Z70.1–1955.

3.2.2.1.1 *Diameter and depth.* — The diameter and depth of the hub cavity shall be inspected as specified in 4.4.2. Hubs which do not fit gages within the limits as specified in 4.4.2 shall be considered not acceptable. In addition, when the hub is fitted onto gages as specified in 4.4.2, there shall be no movement of the hub on the gage when the cannula proper is pressed lightly from side to side in all directions.

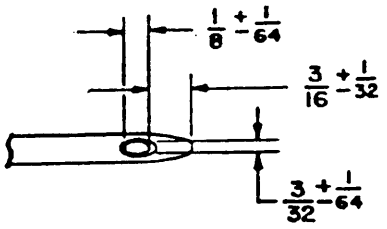
3.2.2.1.2 *Freedom from foreign matter.*—The hub cavity shall be tested for freedom from foreign matter as specified in 4.4.3 and shall show complete freedom from metal chips and other foreign matter.

3.2.2.1.3 *Fluid tightness.*—The hub shall be tested for fluid tightness as specified in 4.4.4. The soldered connections of the cannula proper with the hub shall show no leakage, and the Luer connection of the hub to the plug gage shall show leakage not exceeding the formation of a falling drop when pressure is applied for 10 seconds.

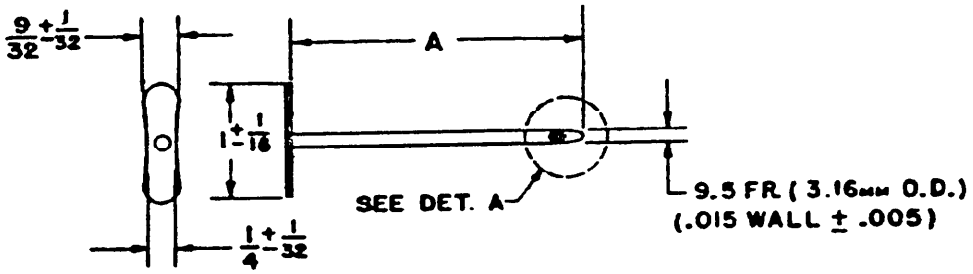
3.2.3 *Type III (suction tube).*—Type III shall be an 8 Fr. suction tube, Frazier design (curved), with finger release valve. The cannula shall be in accordance with figure 3. Cannula shall be supplied with a stylet (see figure 3) of such length that when the stylet is fully inserted into the cannula, the distal end of the stylet shall be flush with the

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SIZE	A-LENGTH (± 1/16)
SMALL	2
MED.	2 1/2
LARGE	2 3/4

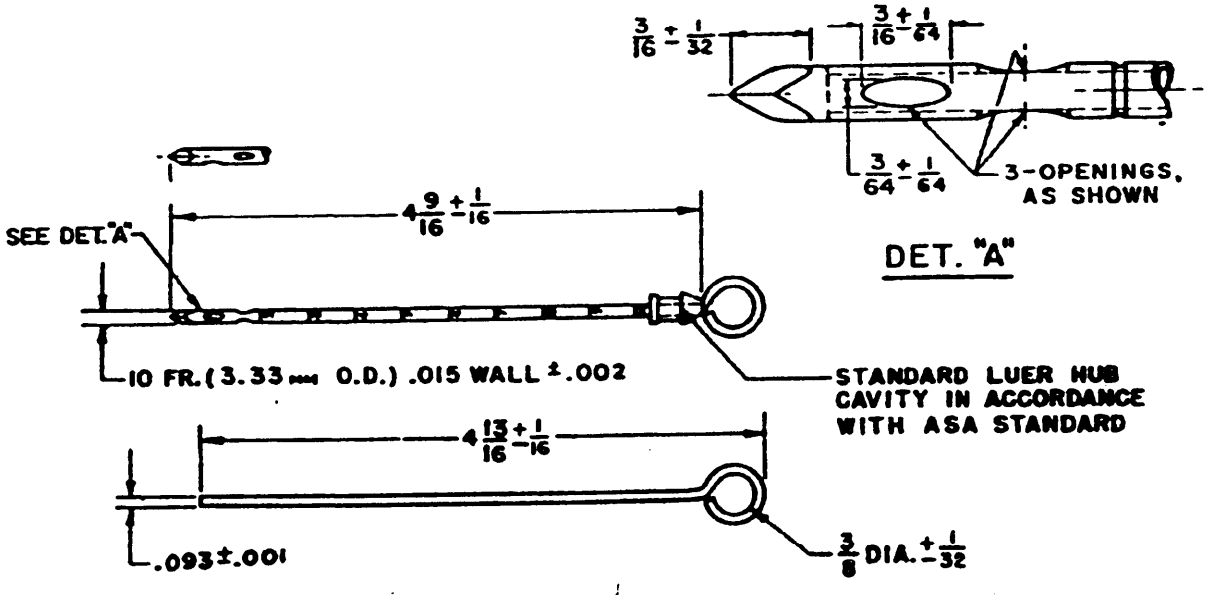


DETAIL A
ENLARGED



TYPE I – EXPLORING

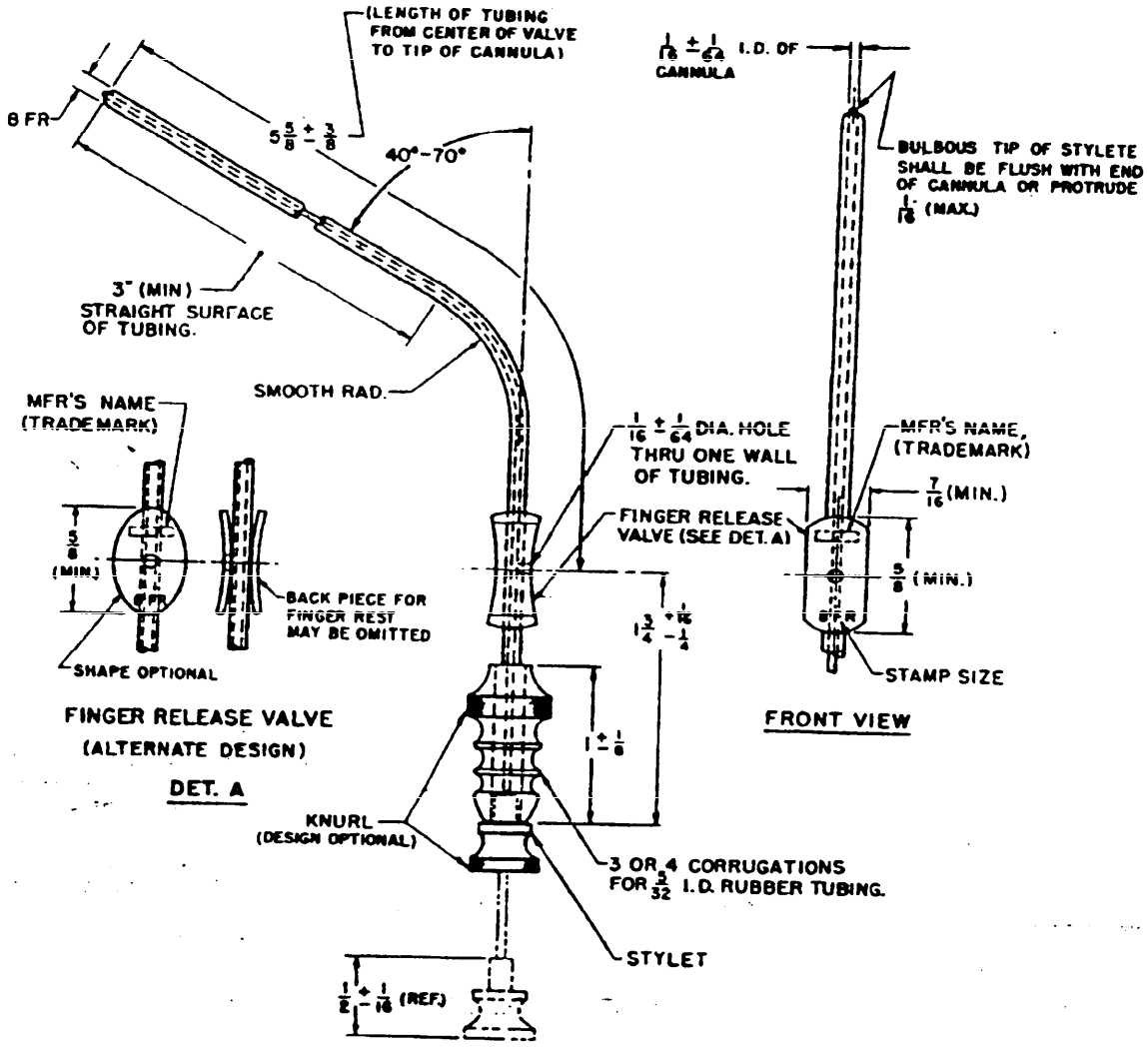
FIGURE 1.



TYPE II – VENTRICULAR

FIGURE 2.

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TYPE III – SUCTION TUBE

FIGURE 3.

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cannula tip or shall protrude slightly beyond the cannula tip up to 1/16 inch). The stylet tip shall be bulbous shaped.

3.3 Security of assembly.—When tested as specified in 4.4.1, the hub, or finger rest, or finger release valve shall not loosen or become detached.

3.4 French scale.—The diameter expressed in millimeters (mm.) according to the French scale is numerically equal to one-third the scale size. A tolerance of plus or minus 1 Fr. (0.3 mm.) is acceptable.

3.5 Finish.—Finish shall be mirror, except type III shall be mirror or satin.

3.6 Additional requirements. — Except as specified herein, the cannula shall comply with requirements of Federal Specification GG–I–526. The requirements of that specification for micrometallographic examination shall not apply.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. —Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and the test results obtained, and indicating that material is as specified, may be furnished.

4.2 Sampling.

4.2.1 *For examination.*—Sampling for examination shall be conducted in accordance with instructions furnished by the procuring activity.

4.2.2 *For tests.*—Sampling for test shall be conducted in accordance with Military Standard MIL–STD–105 and Appendix thereto.

Component	Characteristic	Req.	Test procedure	Unit of product	Reqmt. Appl. Indiv. unit	Insp. level	AQL
Cannula	Security of assembly	3.3	4.4.1	1–Cannula	X	L5	1.5
	Diameter and depth	3.2.2.1.1	4.4.2	1–Cannula	X	II	1.0
	Freedom from foreign matter	3.2.2.1.2	4.4.3	1–Cannula	X	II	2.5
	Fluid tightness	3.2.2.1.3	4.4.4	1–Cannula	X	L5	1.5
	Boil, salt spray and hardness tests	GG–I–526	GG–I–526	1–Cannula	X	L2*	**

*A minimum of two samples shall be tested regardless of lot size.
**Acceptance number shall be zero for all sample sizes.

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4.2.3 *Raw materials.*—Four ounces of each lot of each metal used in the construction of the cannula shall be selected for chemical analysis.

4.2.4 *Components.*—Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the cannula to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more tests characteristics.

4.3 **Examination.**—The cannula shall be examined to determine compliance with all requirements contained in this specification.

4.4 **Tests.** — Tests shall be conducted as necessary, to determine compliance with specification requirements.

4.4.1 *Security of assembly test.*—The cannula shall be securely clamped and the hub, or finger rest, or finger release valve shall be submitted to a pull of 25 pounds.

4.4.2 *Diameter and depth.* — Hub cavity shall be tested on the male Luer taper gage in accordance with plug gage of figure 5 of ASA Standard Z70.1–1955. This gage shall be as shown on the ASA Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.4. The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figure 5 of ASA Standard Z70.1–1955. The hub shall fit the plug gage within the "GO—NO GO" limits of the feeler gage as shown on figure 5 and figure 6 of ASA Standard Z70.1–1955.

4.4.3 *Freedom from foreign matter.*—The hub cavity shall be tested for freedom from foreign matter by visual examination with a magnifier and good illumination and shall also show complete absence of metal chips and other foreign matter upon visual examination under a magnifier of the cotton fibers of a cotton swab (tightly wrapped on the end of small diameter wooden applicator) which has been vigorously turned down into the bottom of the hub cavity.

4.4.4 *Fluid tightness.*—With the cannula in a horizontal position, the hub attached by means of plug gage of figure 5 of ASA Standard Z70.1–1955 (except that gage shall have a fluid path) to a water pressure source having a pressure gage, the water shall be run through the cannula so as to fill any hub cavity space and the lumen, the lumen then sealed by a piece of solid rubber slipped onto the cannula point and a pressure of 45 pounds per square inch applied for 10 seconds.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing used are defined in Federal Standard No. 102.

5.1 **Packaging.**

5.1.1 *Unit of issue.*—Each cannula or set of the stock number specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is

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TABLE I.

Stock No.	Unit of issue	Number of unit packages to be contained in each intermediate package	Number of intermediate packages to be contained in each exterior container	Number of units of issue in each exterior container
6515–813–8800 ------	Set	12	36	432
6515–313–8840 ------	Each	12	36	432
6515–386–8600 ------	Each	6	48	288

less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*

5.1.3.1.1 *Cleaning.* — Each unit shall be thoroughly cleaned in accordance with the applicable method as outlined in Military Specification MIL–P–116.

5.1.3.1.2 *Wrapping and cartoning.*—Each unit shall be completely wrapped in neutral tissue or other suitable anti-corrosive material and then packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.3.1.3 *Bag.*—Each unit carton shall be further packaged in a bag conforming to Military Specification MIL–B–117, type II, class d.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level B.*

5.1.4.1 *Unit package.*

5.1.4.1.1 *Cleaning.*—Cleaning shall be as specified for level A.

5.1.4.1.2 *Wrapping and cartoning.*—Wrapping and cartoning shall be as specified for level A.

5.1.4.1.3 *Bag.*—Bag will not be required for level A packages.

5.1.4.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size, or a complete wrapping of kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.5 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be conducted in accordance with Federal Specification PPP–B–621, table III.

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5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Interior packages—military.* — Each unit carton and bag, and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Exterior container—military.* — Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2(c), 6.2(d), 6.5, and 7.5.1 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

5.3.3 *Exterior container—nonmilitary.* — In addition to any special markings specified in the contract or order, exterior containers shall be marked in accordance with Federal Standard No. 123.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Types required (see 1.2).

(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(1) Level B packaging is intended to provide economical but limited protection, and should be specified only when it is determined the cannulae will be held in covered storage not more than 12 months from date of initial packaging.

6.2 This specification covers the following items appearing in the Federal Supply Catalog, Department of Defense Section, for Medical Materiel:

6515–313–8800 CANNULA SET, BRAIN, Frazier.

6515–313–8840 CANNULA, BRAIN, Frazier, 10 Fr.

6515–386–6800 CANNULA, BRAIN, Frazier, 8 Fr.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 **Transportation description.**—Transportation description and minimum weight applicable to this commodity are:

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Rail:

Surgical instruments, not otherwise indexed by name.

Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Surgical instruments, not otherwise indexed.

Truckload minimum weight 10,000 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 10 cents each.

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PLATE NO. 15411

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 194

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